Bluray Drive used: The ASUS BW-16D1X-U External Blu-ray Drive

OS Used: Linux, MacOS and Windows 11

I have my Movie/TV Show Library stored on my UNAS Pro, which is then connected to my Jellyfin Server.

I originally used the jellyfin server client on Ubuntu but switched to Windows.

ISP and Speeds: Spectrum Fiber (1000+mbps Up and 1000+mbps Down, over ethernet consistently)

1. Download **MakeMKV** and **SFD Tool** as well as **“The all you need firmware”** pack from the MakeMKV Website.
2. In order to be able to rip 4K UHD Discs you will need to flash the drive to the **ASUS-BW-16D1HT-3.10-WM01601.bin** file
3. After flashing the Blueray player you will be able to rip 4K UHD Discs, if you do not need to rip 4K UHD Discs you will not need to flash the player with this firmware. The player will by default play Bluray discs at 1080p.
4. When flashing the bluray player with SFD tool it is a .exe file, I was only able to do this on a windows computer, you may be able to on mac or linux, However i am unsure of this.

Jellyfin Documentation / Transcoding Information

Originally I used a Dell Optiplex 7050 MFF with an I5 - 6500t, which is a low power cpu. This CPU is capable of running a Jellyfin server however it is not capable of transcoding in 4K UHD, it puts too much strain on the CPU itself.

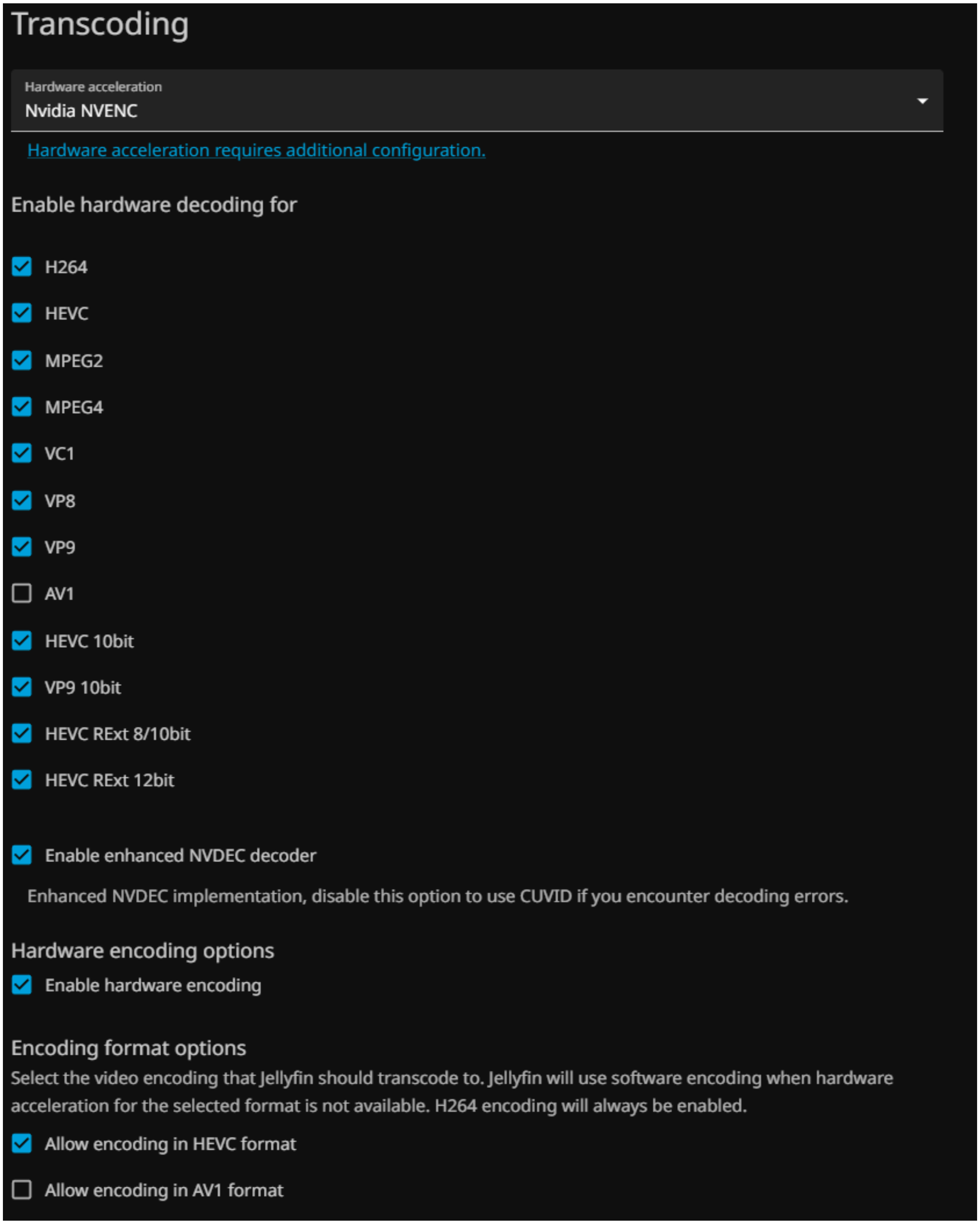
In order to solve this I would need a more powerful PC to transcode in 4k, preferably one with a desktop GPU. With some old parts laying around i put together a Jellyfin server the specs are as follows:

AMD 3600x
8GB 2400mhz RAM (2133 MHZ)
GTX 1070
GTX 960 (The GTX 960 is not being used by jellyfin, it is used by another application)

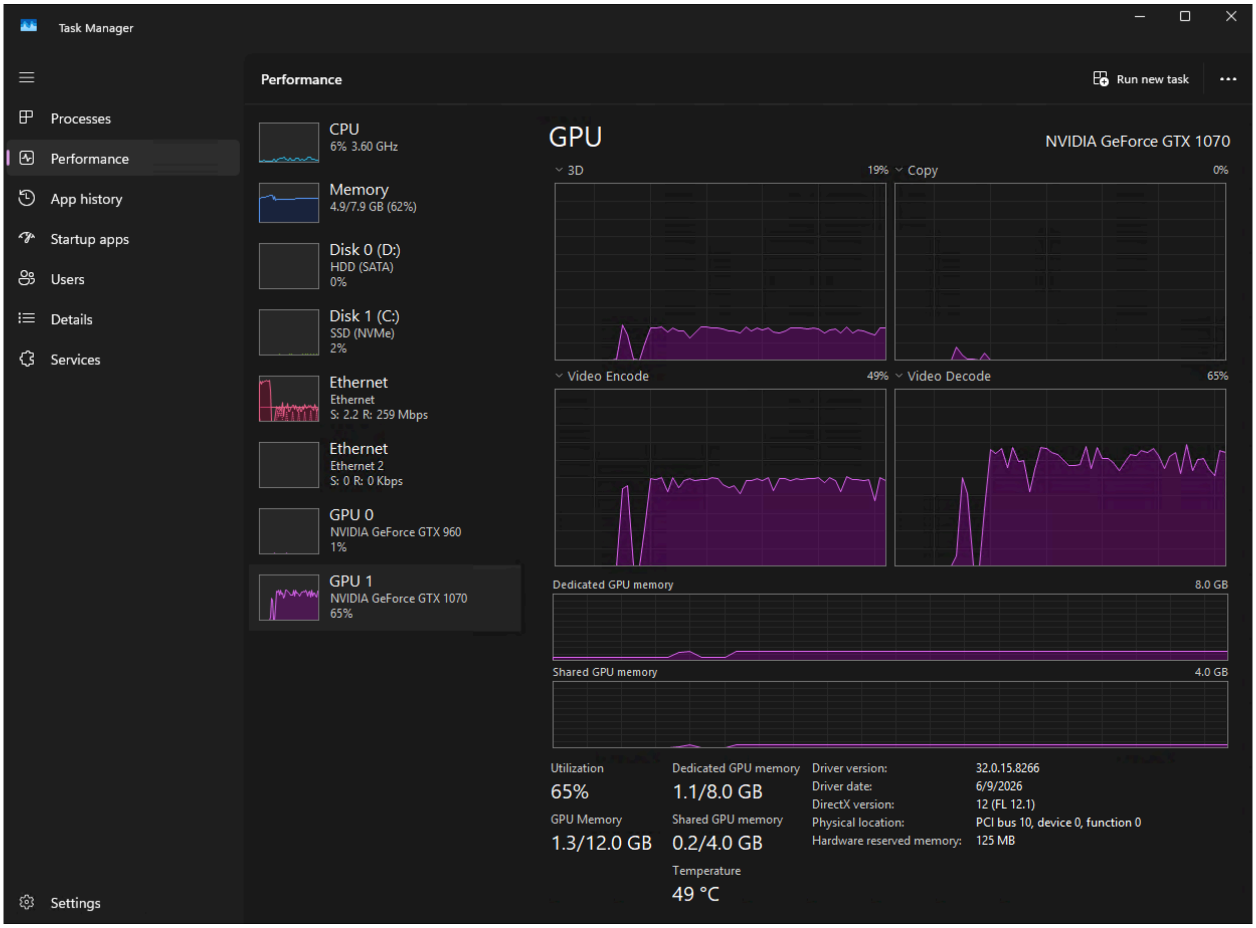
I built this computer and had an unused m.2 ssd laying around that already had windows 11 installed and a profile so I just stuck with that. I now have jellyfin running off of it.

My issue with this was having the server transcode 4K UHD using the server GPU (GTX 1070).

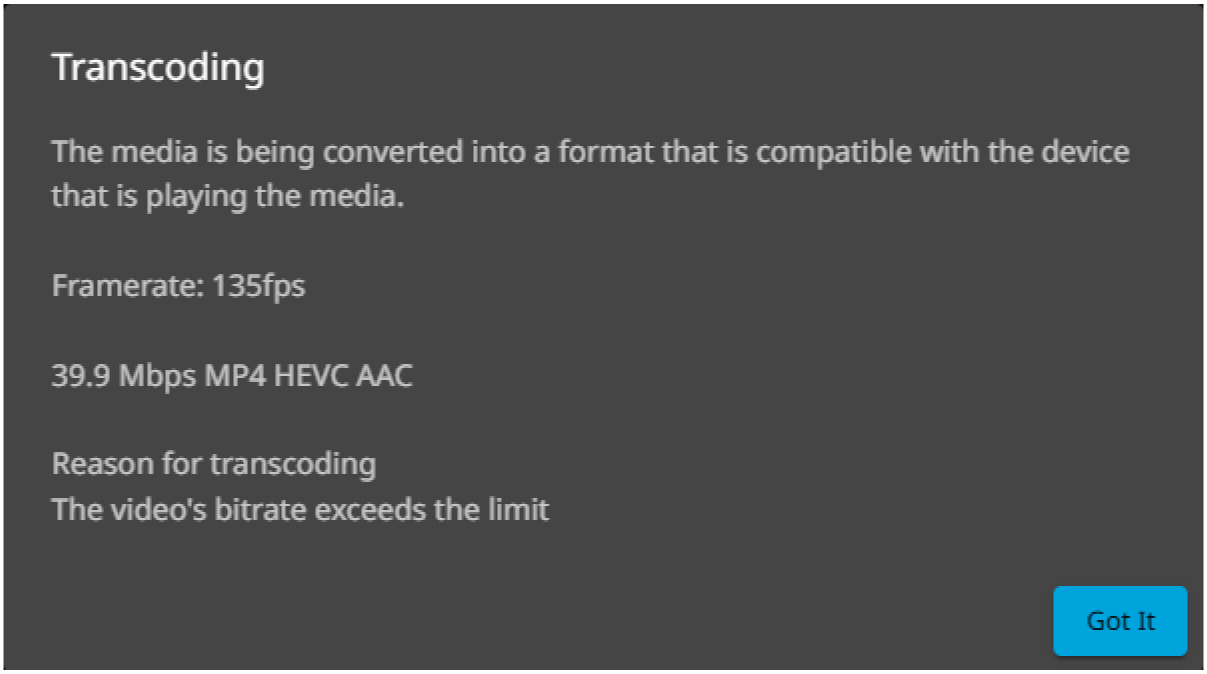
The GTX 1070 does not support AV1 encoding. AV1 NVENC requires an RTX 4000 series or newer. Jellyfin is trying to transcode to AV1 and the 1070 can't do it.
I wanted my client PC to use the web client and transcode using the server GPU (GTX 1070). In order to do this I had to enable NVIDIA NVENC and Disable AV1 as well as Allow encoding in AV1 format. Here is an image:



Here is a result of doing this with an NVIDIA Card:



The GPU is now being used to transcode. Here is the Info from the Jellyfin Dashboard:



Some things to note: NVIDIA NVENC transcoding does not work on MacOS.

After doing some testing the Jellfin Application you can get from the app store on the Mac does not really work for transcoding while using a NVIDIA GPU.

If you are using a web browser on a Mac, it will not use the server GPU to transcode. It will use local resources/Hardware to transcode. This is because MacOS does not support NVIDIA transcoding.

Sony TV (Android TV) and Jellyfin

Very simple setup, just download the app from the playstore, enter the ip address of the server, login with a user profile and everything should be there. In my experience if your tv and server are on the same network / VLAN, the TV may even auto detect the server.

Using the Sony TV it is a Direct Play of the RAW File from my UNAS Pro that has the file. (TV Is on ethernet)

Should say something like this for a file directly playing.